

Project Proposal

Project Title: Identifying Different Types of Brain Tumors Using Deep Learning

The data set I am using is called '**Brain Tumor MRI Datasets**' published on Kaggle by MASOUD NICKPARVAR

The link to the dataset: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset/data>

This dataset contains 4 groups; 3 types of brain tumors (glioma, meningioma, and pituitary tumors) and the control group (samples with no tumor)

This dataset contains 7023 images, 5712 for training and 1311 for testing (each category has approximately the same number of images)

This dataset fits for my project because I can use the images from this dataset to classify three different types of tumors based on the tumor's location.

The algorithm I hope to use is the CNN network built by Keras, I also intend to use the LSTM algorithm to see if I can improve my results. I will evaluate the results using the metric 'accuracy'

The main challenge I can already see is that the files are quite large, which means it would take a long time to train. Furthermore, some images are cross-sections, while others are long-sections. I do not know if I should split the these 2 types of data (it would take a long time) or if there is a simpler approach.